MACHINE LEARNING FUNDAMENTALS

Multiclass Classification vs. Multi-Label Classification



MULTICLASS



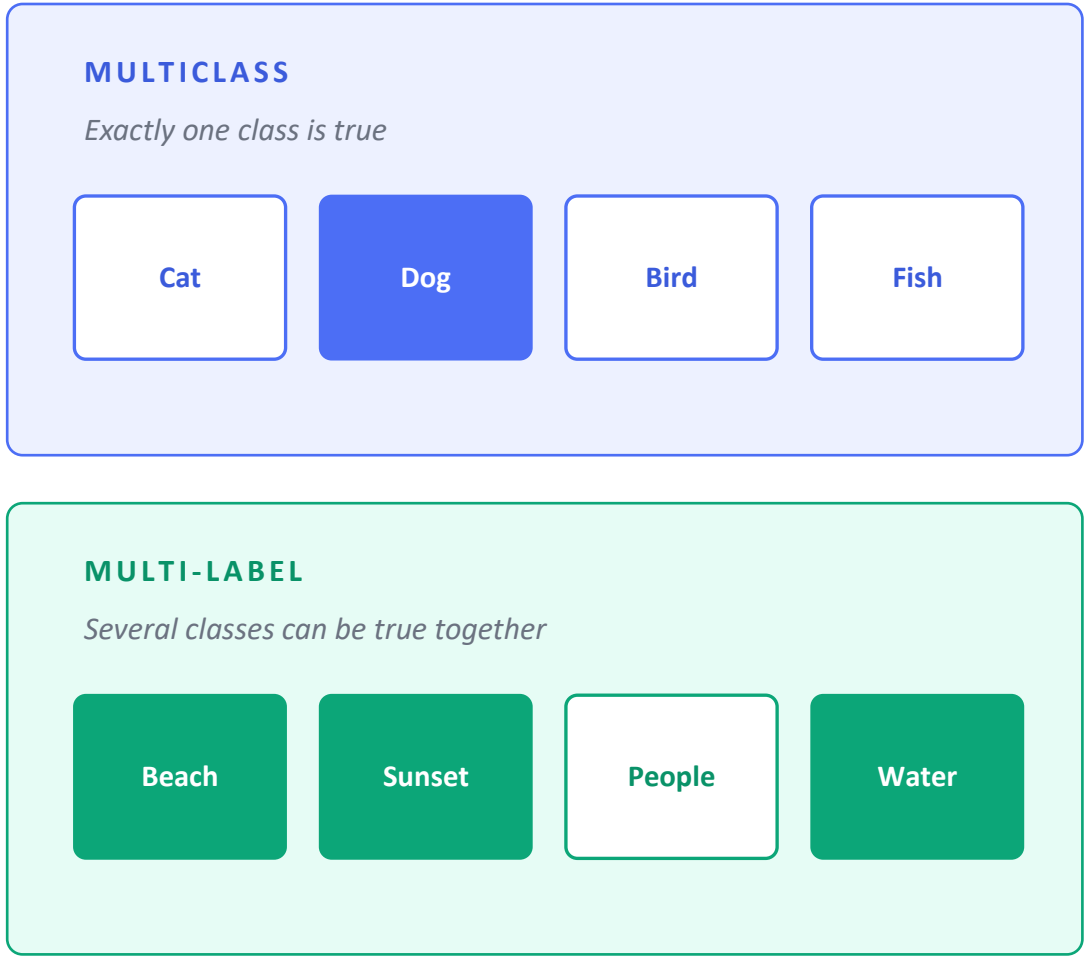
MULTI-LABEL

How many labels can one input truly have — and why that single question changes everything downstream

INTRODUCTION

Two Ways to Assign Labels

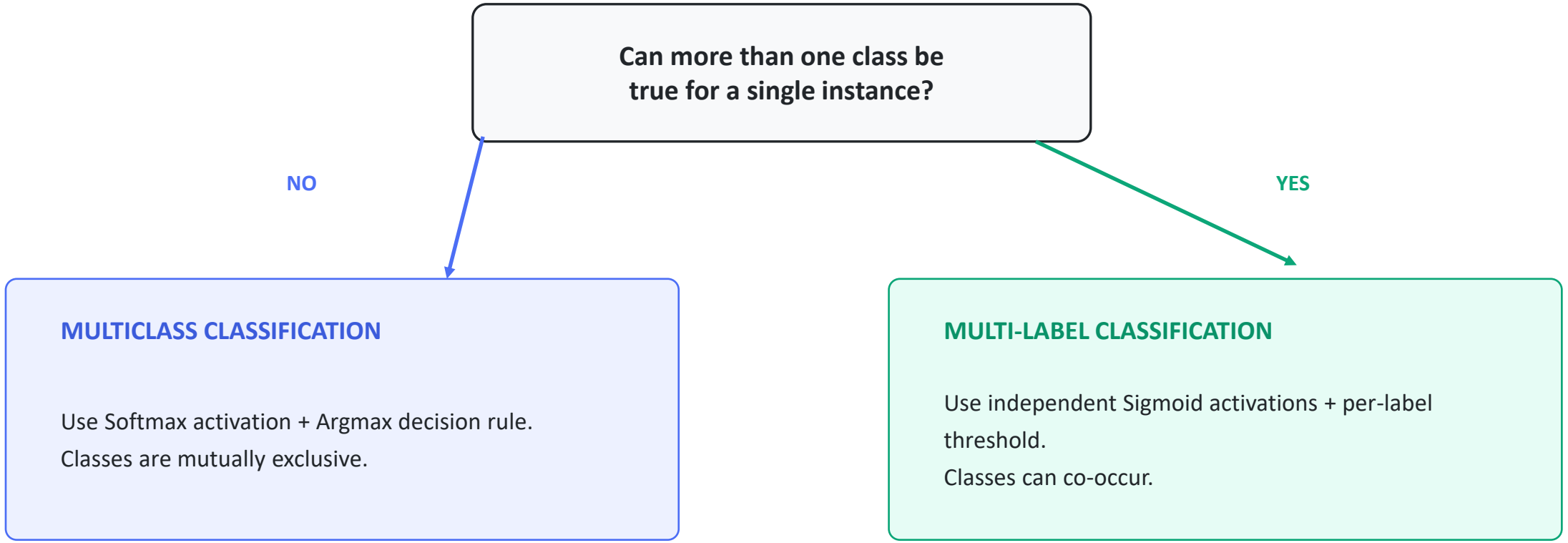
- Classification tasks differ fundamentally in how many labels can be true for a single input at once.
- Multiclass classification assigns exactly one label out of several mutually exclusive classes.
- Multi-label classification allows an input to carry any number of relevant labels at the same time — zero, one, or several.
- This one distinction cascades through architecture, activation functions, loss functions, and evaluation metrics.



FRAMING THE PROBLEM

Choosing the Right Framework

Everything starts with one question about the data itself — not the model.



Getting this decision wrong silently corrupts the loss function and every metric computed afterward — even while the model still "trains."

DEEP DIVE

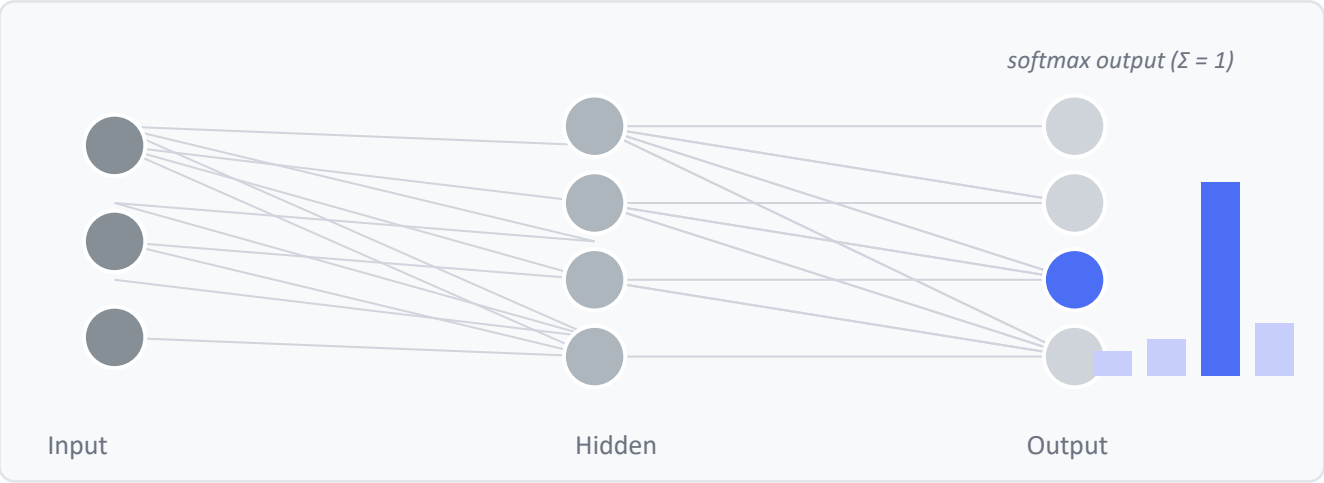
Multiclass Classification: One Input, One True Class

- Assigns each input exactly one label from a set of N ≥ 3 mutually exclusive categories.
- Classes are exhaustive and non-overlapping — a fruit is an apple OR a banana OR an orange, never more than one at a time.
- The model partitions the entire feature space into N non-overlapping decision regions.
- The output represents a single probability distribution over all classes that sums to 1.
- Own insight: multiclass problems are natural whenever categories are exclusive by definition — a coin can only land on one face.



DEEP DIVE

Multiclass Architecture & Activation



- **Activation:** Softmax converts raw output logits into a single probability distribution across all classes.
- **Loss function**: Categorical Cross-Entropy penalizes divergence from the one-hot encoded true label.
- **Prediction rule:** Argmax — the class with the single highest probability becomes the final prediction.
- Own insight: because probabilities must sum to 1, gaining confidence in one class necessarily steals it from every other — classes compete by design.

MULTICLASS IN PRACTICE

Real-World Examples

1

Handwritten Digit Recognition

Classifying an image as exactly one digit, 0 through 9 (e.g., MNIST).

2

Iris Flower Species

Assigning a flower to Setosa, Versicolor, or Virginica from petal/sepal measurements.

3

News Article Categorization

Sorting an article into a single section — Sports, Politics, or Technology.

4

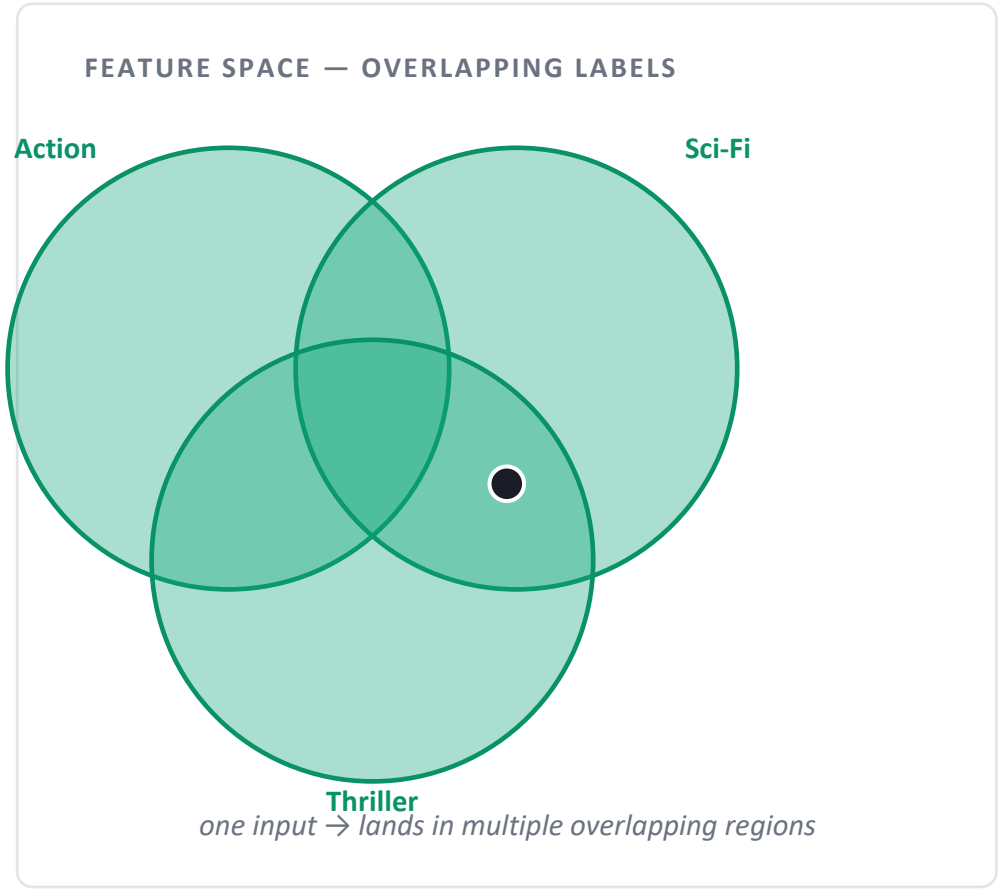
Sentiment Analysis (3-class)

Labeling a review as Positive, Neutral, or Negative — one sentiment per review.

DEEP DIVE

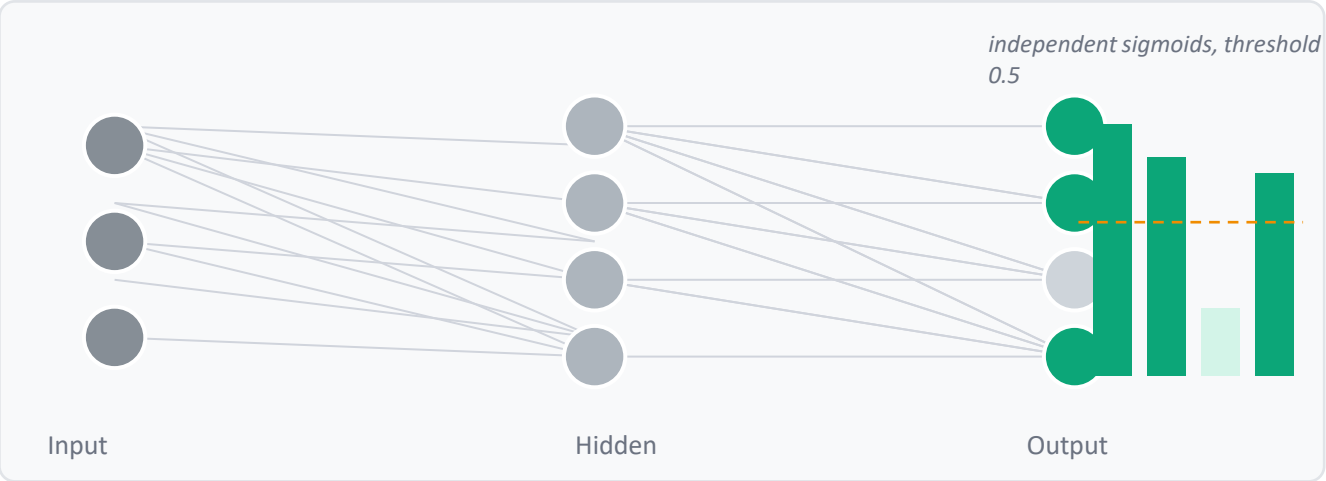
Multi-Label Classification: One Input, Many Possible Classes

- Assigns each input any combination of labels from a set of categories — zero, one, or many can apply at once.
- Each label is treated as an independent binary decision rather than one joint decision across all classes.
- Real-world data is rarely mutually exclusive — objects, texts, and patients routinely carry multiple valid attributes simultaneously.
- Own insight: model complexity grows because label co-occurrence and correlation patterns must be learned, not just clean class boundaries.



DEEP DIVE

Multi-Label Architecture & Activation



- Activation: each output neuron uses an independent Sigmoid, producing its own probability between 0 and 1.
- Loss function: Binary Cross-Entropy is computed per label, then summed or averaged across all labels.
- Prediction rule: apply a threshold (commonly 0.5) per label — any label crossing it joins the final prediction set.
- Own insight: because each label is decided independently, thresholds can be tuned per label to balance precision and recall for that specific class.

MULTI-LABEL IN PRACTICE

Real-World Examples

1 Movie Genres

A movie tagged as Action, Sci-Fi, AND Thriller.

2 Image Annotation

Detecting a Person, Car, AND Dog in one photo.

3 Medical Diagnosis

A patient presenting Hypertension AND Diabetes.

4 Music Tagging

ADDED

A song labeled “Chill,” “Acoustic,” AND “Instrumental” at the same time.

COMPARISON

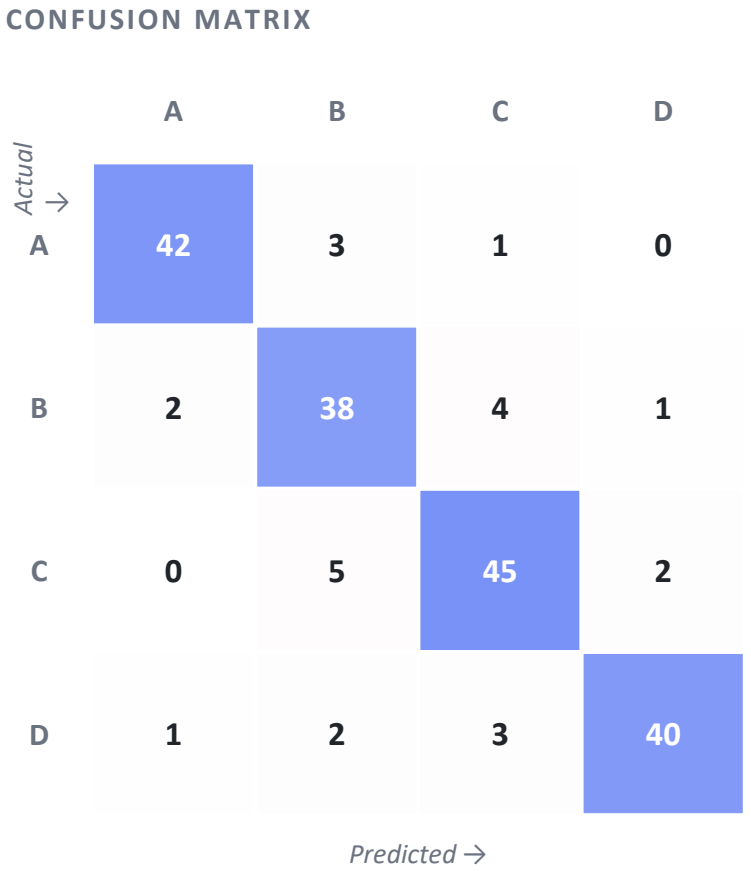
Side-by-Side Architectural Comparison

DIMENSION	MULTICLASS	MULTI-LABEL
Output activation	Softmax (joint distribution)	Sigmoid (per label, independent)
Loss function	Categorical Cross-Entropy	Binary Cross-Entropy (per label)
Label relationship	Mutually exclusive	Independent / co-occurring
Prediction rule	Argmax (top-1 class)	Per-label threshold (e.g., 0.5)
Output constraint	Probabilities sum to 1	No sum constraint

EVALUATION

Measuring Multiclass Performance

- **Accuracy**: fraction of instances where the predicted class exactly matches the true class.
- Confusion Matrix: an N×N grid of true vs. predicted classes that reveals which classes get confused with one another.
- **Precision, Recall, F1 (per class):** standard binary metrics computed for each class individually, then aggregated.
- **Macro vs. Micro averaging:** macro treats every class equally; micro weights by class frequency — better for imbalanced data.

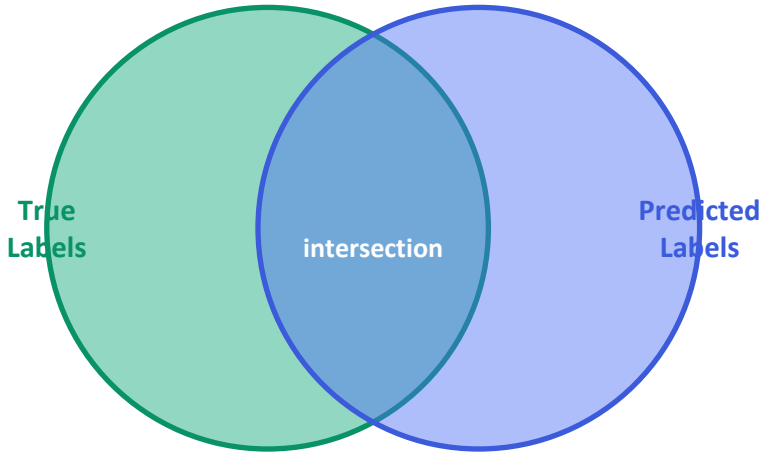


EVALUATION

Measuring Multi-Label Performance

- **Hamming Loss**: the fraction of individual labels predicted incorrectly (wrong presence or absence), averaged over all labels and samples.
- **Subset Accuracy (Exact Match Ratio)**: requires the entire predicted label set to exactly match the true set — a strict, often harsh metric.
- Micro / Macro F1: the same averaging concept as multiclass, but computed across the full set of binary label decisions.
- Jaccard Index: intersection over union of predicted and true label sets; rewards partial correctness that exact-match metrics ignore.

JACCARD INDEX = |Predicted ∩ True| / |Predicted ∪ True|



A higher Jaccard score means the predicted set overlaps more fully with the true set of labels — partial credit that Hamming Loss and Exact Match don't give.

TRADE-OFFS

Pros, Cons & Trade-offs

MULTICLASS

ADVANTAGES

Simpler model setup, lower computational cost, straightforward interpretation and standard metric setup.

DISADVANTAGES

Cannot handle real-world overlapping categories; forces artificial boundaries if data naturally contains co-occurring attributes.

MULTI-LABEL

ADVANTAGES

Captures complex, real-world context and co-occurring features.

DISADVANTAGES

Higher computational complexity, challenging label correlations, requires complex metrics to measure partial correctness.